

Which Computation Answers the Physics Question?

- Start with the required physical output
- Choose among methods already learned
- Finish with a defensible capstone result

1. Physical Question

What is the temperature $T(t)$ of the body over time?



Newton cooling model

$$\frac{dT}{dt} = -\frac{(T - T_{\text{env}})}{\tau}$$

$T_{\text{env}} = 20\text{ }^{\circ}\text{C}$

$T_0 = 80\text{ }^{\circ}\text{C}$ at $t = 0\text{ s}$

$\tau = 100\text{ s}$

$0 \leq t \leq 200\text{ s}$

Required physical output:

$T(t = 200\text{ s})$ in $^{\circ}\text{C}$

2. Computation

Use a method already learned and within its limitation



Euler method

- $\Delta t = 20\text{ s}$
- $\Delta t = 10\text{ s}$

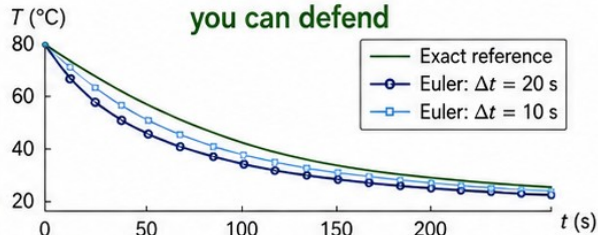


Exact solution (reference)

Check a limitation you can justify (step size, domain, assumptions)

3. Evidence

Create and check evidence you can defend



Method	$T(200\text{ s})$ ($^{\circ}\text{C}$)	Absolute error ($^{\circ}\text{C}$)
Exact (reference)	28.1201	—
Euler, $\Delta t = 20\text{ s}$	26.4425	1.6777
Euler, $\Delta t = 10\text{ s}$	27.2946	0.8255

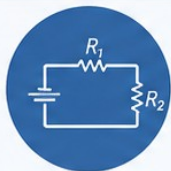
Document the required check, your chosen check, limitation and rationale

One question. Known methods. | Careful computation. | Defensible evidence.

Name the Output Before Choosing the Method

Start with the question. Let the output you want guide the method you use.

What are you trying to find?



What are the currents in each branch of this circuit?
(several unknowns at once)



Linear system

Set up linear equations and solve.
(e.g., MATLAB backslash `\`)



At what position does the ball's height first reach zero?
(find the root of a function)



Root finding

Solve $f(x) = 0$ for x .
(e.g., MATLAB `fzero`)



What is the total work done from a to b ?
(accumulated amount over an interval)



Integration

Accumulate area under a curve.
(e.g., MATLAB `integral` or `trapz`)



How does the temperature change over time?
(state evolves with time)

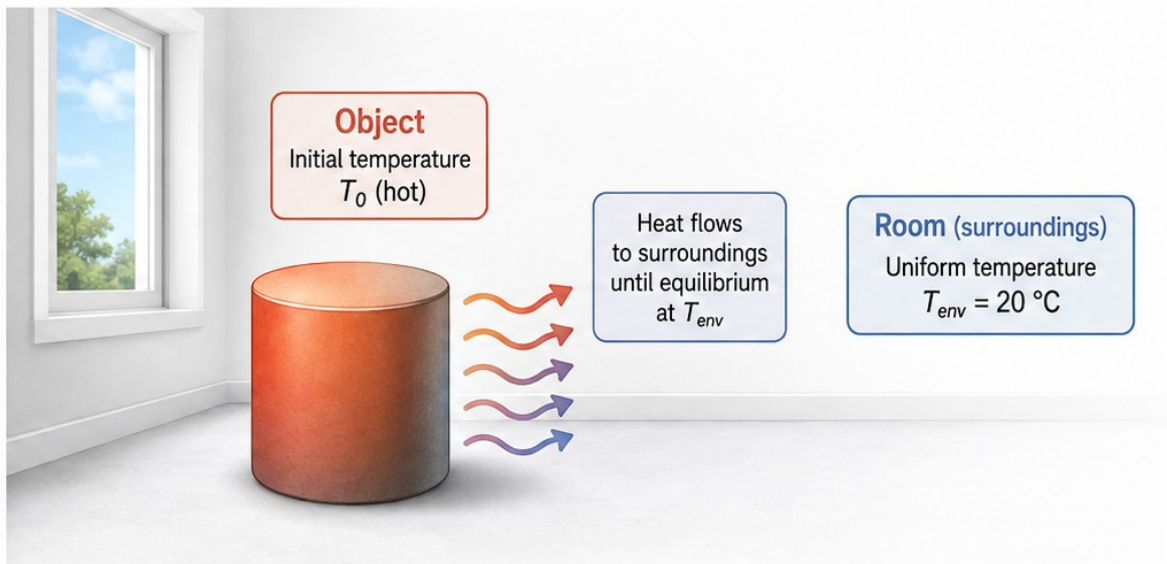


ODE stepping

Step an initial-value ODE forward in time.
(e.g., Euler, `ode45`, custom stepper)

Name the output first → match the method → check the limitation.

Inputs, Output, and One Limitation



Inputs:

T_0 , T_{env} , τ and times



Output:

temperature versus time



Assumptions:

uniform object and
constant surroundings



Limitation:

finite timestep and
model assumptions



Integrated Method Recognition

Identify and connect
the method inside
the model



Supplied-Code Auditing

Read, trace, and verify
the implementation



Validation

Check results against the
locked reference to confirm
the implementation



Formative Capstone Evidence Assembly

Document, reflect,
and assemble your
evidence

Predict the Cooling Before Running MATLAB

Use the model first. Do not wait for the plot.

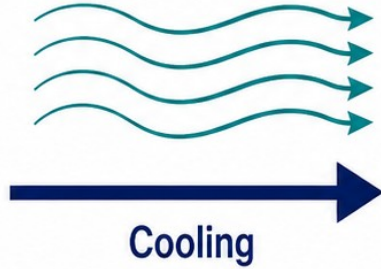
Initial state ($t = 0$ s)



$T = 80\text{ °C}$

System hotter than
surroundings

Heat flows to
cool surroundings (20 °C)



Later state ($t > 0$)



T moves toward
 20 °C

Temperature
decreases

Prediction Checklist

- ✓ Start hotter than the surroundings
- ✓ Decrease toward 20 °C
- ✓ Initial value is 80 °C
- ✓ A rising curve contradicts this model



Prediction: Temperature must decrease from 80 °C toward 20 °C over time.

If your output shows a rise, pause. Re-check the model, sign, and first update.

Connect the Rate to the Update

Newton
Cooling Law

$$\frac{dT}{dt} = -\frac{(T - T_{\text{env}})}{\tau}$$



Rate: $-(T - T_{\text{env}})/\tau$
Rate units: deg C per second

Euler
Update

$$T_{\text{next}} = T_{\text{current}} + dt * \left[-\frac{(T_{\text{current}} - T_{\text{env}})}{\tau} \right]$$

$\overbrace{\text{deg C / s}}$

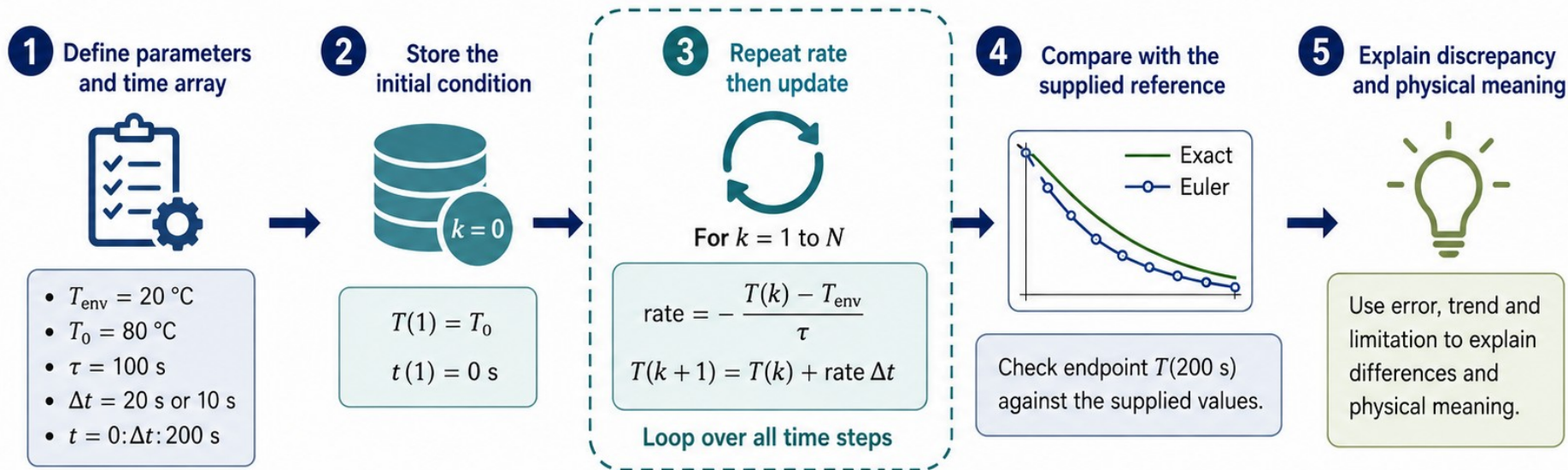


deg C / s * s = deg C
 $\underbrace{\hspace{1cm}}_{dt \text{ units}} \quad \underbrace{\hspace{1cm}}_{\text{temperature change}}$

- 1 Multiply by dt to get a temperature change
- 2 Add the change to the **current** state

Read the Algorithm Before the Code

Make the computational recipe explicit before you write or audit MATLAB.



- 1 Define parameters and time array** — Set T_{env} , T_0 , τ , Δt and the time vector t .
- 2 Store the initial condition** — Record the known starting temperature at $t = 0$.
- 3 Repeat rate then update** — Compute the rate, update T , and advance the loop over all steps.
- 4 Compare with the supplied reference** — Validate against the exact or higher-fidelity reference.
- 5 Explain discrepancy and physical meaning** — Interpret error, trend and limitation in context.

Trace One Euler Step

MATLAB code:

```
T_C(1)=T0;  
dTdt=-(T_C(1)-Tenv)/tau;  
T_C(2)=T_C(1)+dt*dTdt;
```

Given values:

$T_0 = 80$, $T_{env} = 20$, $\tau = 100$ s, $dt = 20$ s

One Euler step calculation:



$T_C(1)=80$ at $t=0$. After one Euler step with $dt=20$ s, the first updated value is $T_C(2)=68$ deg C at $t=20$ s.

Trace of T_C through one Euler step

index	time	temperature
1	0 s	80 deg C
2	20 s	68 deg C

Index 1 represents $t = 0$ (initial condition).
Index 2 is the first updated value at 20 s.

Key points / trace facts:

- $T(1)=80$ at time zero
- $dt=20$ s
- Initial rate= -0.6 deg C/s
- Change= -12 deg C
- $T(2)=68$ deg C at 20 s

A Script Can Run and Still Be Wrong

DEFECTIVE

```
% Newton cooling (DEFECTIVE SIGN)
```

```
Tenv = 20;
```

```
T = 80;
```

```
tau = 100;
```

```
dt = 20;
```

```
dT = + (T - Tenv)/tau;    % WRONG sign
```

```
Tnext = T + dt*dT;
```

Locked model facts

$$\text{Newton cooling } \frac{dT}{dt} = - \frac{(T - T_{\text{env}})}{\tau}$$

$$T_{\text{env}} = 20 \text{ deg C}$$

$$T_0 = 80 \text{ deg C}$$

$$\tau = 100 \text{ s}$$

$$\text{Euler dt} = 20 \text{ s}$$

REPAIR

```
% Newton cooling (CORRECT SIGN)
```

```
Tenv = 20;
```

```
T = 80;
```

```
tau = 100;
```

```
dt = 20;
```

```
dT = - (T - Tenv)/tau;    % CORRECT sign
```

```
Tnext = T + dt*dT;
```

- 1 Plus-sign rate gives 92 deg C on the first step

$$dT = + (80 - 20)/100 = +0.6 \text{ deg C/s}$$

$$T_{\text{next}} = 80 + 20(0.6) = 92 \text{ deg C}$$

- 2 The result moves away from the room temperature



- 1 Repair the sign in the rate
- 2 Rerun and validate independently

- 1 Minus-sign rate gives 68 deg C on the first step

$$dT = - (80 - 20)/100 = -0.6 \text{ deg C/s}$$

$$T_{\text{next}} = 80 + 20(-0.6) = 68 \text{ deg C}$$

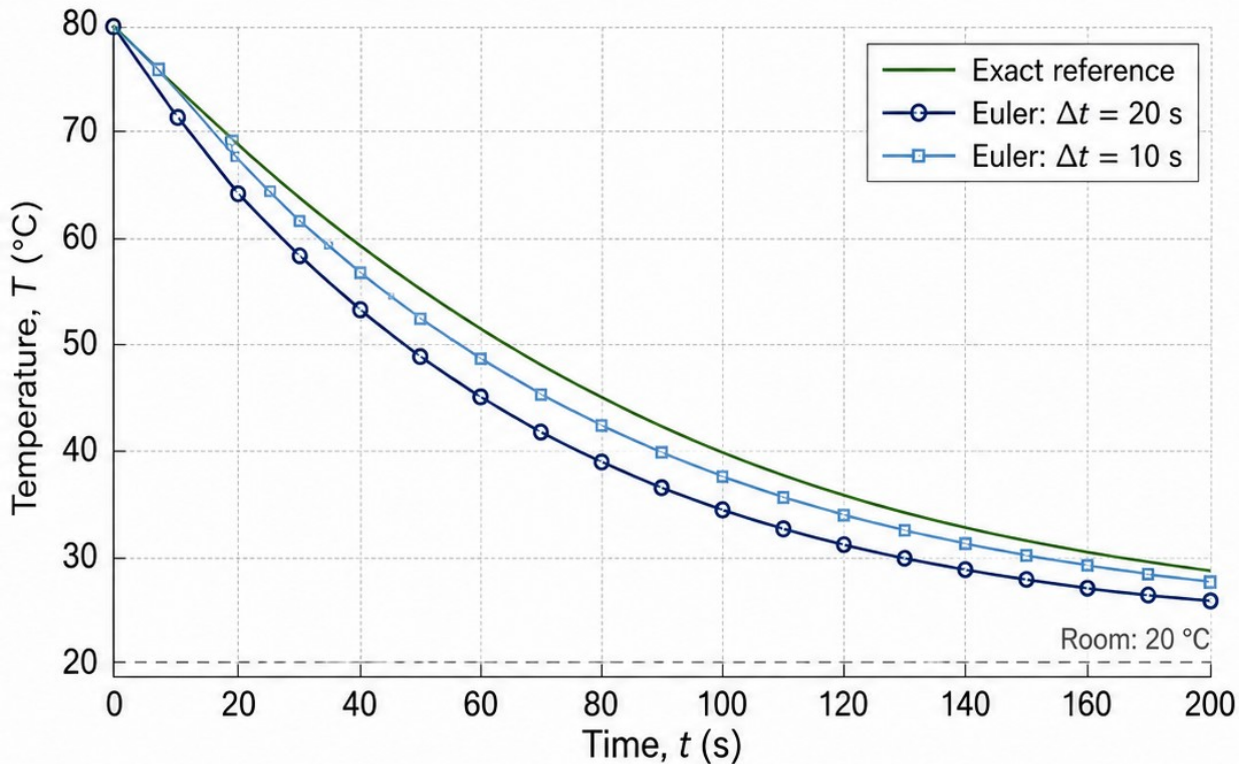
- 2 The result moves toward the room temperature



Syntactically valid code can still be physically wrong. Fix the model sign and validate independently.

Validate Against the Supplied Reference

Same model, same final time: 200 s



At $t = 200$ s

Exact reference
28.1201 °C

Euler step	Temperature	Absolute error
20 s	26.4425 °C	1.6777 °C
10 s	27.2946 °C	0.8255 °C

The finer step is closer to the supplied reference.

This checks the numerical result for the stated model—not every real-world assumption.

Choose a Check That Tests the Claim

Pick the validation check whose purpose matches the claim you want to test.



Initial condition tests setup

Confirms the model starts from the intended state and parameters.



Reference comparison tests numerical output

Compares the numerical result to a trusted reference at the same time.



Timestep refinement tests discretisation sensitivity

Repeats the calculation with smaller steps to see if the result changes systematically.



Units alone cannot catch every sign defect

A result can have correct units yet be wrong due to a sign error.

Example: aligning a claim with the right check



Claim:
Euler cooling is close enough



Check:
compare with supplied exact endpoint

A clear **claim** leads to the right **check**—and a meaningful result.

Change One Parameter and Explain the Physics

What we will do

- Change τ from 100 s to 150 s
- Hold other inputs fixed
- Predict slower cooling
- Compare with the corresponding exact reference

Parameter	Symbol (units)	Value (held fixed)
Initial temperature	T_0 (°C)	80 °C
Ambient temperature	T_∞ (°C)	20 °C
Time step	Δt (s)	20 s
Total time	t_{final} (s)	200 s
Time constant (changed)	τ (s)	100 s → 150 s



All inputs other than τ are fixed.

Controlled change

BEFORE

$\tau = 100$ s

All other inputs fixed

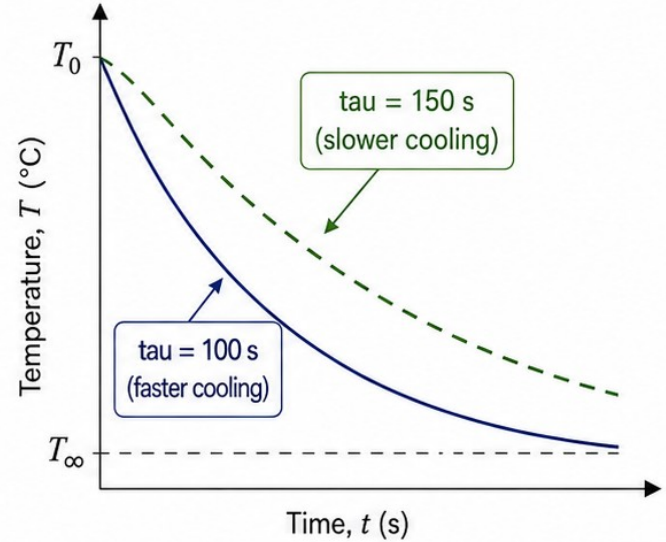


AFTER

$\tau = 150$ s

All other inputs fixed

Conceptual cooling response



Larger τ means a slower approach to ambient temperature.

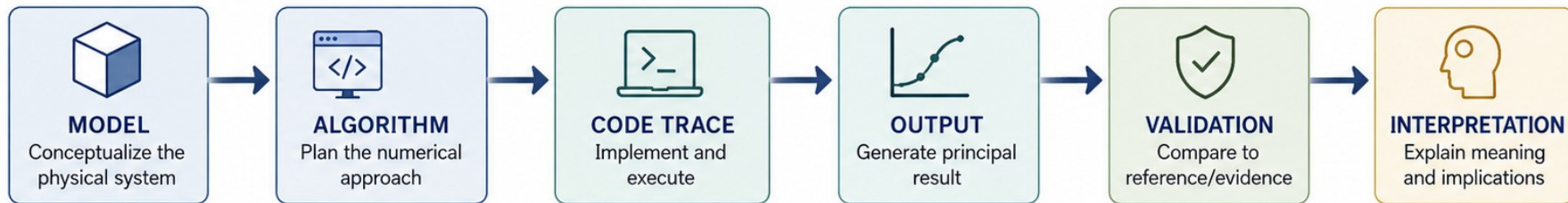


The lecturer **MATLAB** script computes and checks this modification.

It generates the numerical result for **$\tau = 150$ s** and compares it with the corresponding **exact reference**.

PHY4605 Week 12: Assemble the Capstone Evidence

Build and document the complete physical-model-to-interpretation chain.



	1 State model, assumptions, variables and units	REQUIRED	Clearly describe the physical model and assumptions. List all variables with their symbols and units.
	2 Show pseudocode and one justified modification	REQUIRED	Provide pseudocode for your algorithm. Name one modification and justify why you made it.
	3 Include one principal output	REQUIRED	Show your primary result with proper label, units, and a concise caption.
	4 Include required plus chosen validation	REQUIRED CHOSEN	Perform the required validation. Add one additional, chosen validation and report it.
	5 Explain one limitation	REQUIRED	Identify one important limitation of your model, method, or data, and explain its impact.
	6 Provide complete evidence package	CHOSEN	Submit all source files, data, and outputs used to produce your results.
	7 Document key decisions	CHOSEN	Summarize key modeling and coding decisions and how they affect the results.
	8 State next steps	CHOSEN	Outline one improvement or extension you would pursue next and why.

Reproduce and Rehearse the Defence

Fresh-Session Reproducibility Checklist



Rerun from a fresh MATLAB session



Record release, parameters and numerical settings



Declare accepted, modified and rejected AI assistance



Each member traces code and explains the physics

Two/Three-Person Individual Defence Rehearsal Path



1. Individual Run-Through

Each member presents their part end-to-end.



2. Cross-Check & Challenge

Peers ask probing questions, challenge assumptions and verify results.



3. Full Defence Simulation

Take turns as presenter, questioners and moderator. Refine until answers are crisp, evidence-backed and consistent.

What Makes Your Result Trustworthy?



METHOD

1. Name one method from its required output



CHECK

2. Identify one code step and validation result



NEXT ACTION

3. State the capstone item still needing attention